

Bridging Geometric Learning and Pattern Decoupling for Mixed-Type Wafer Defect Recognition via Dual-Collaborative Graph Learning

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Abstract—Wafer defect pattern recognition is critical for defect diagnosis and yield improvement in semiconductor manufacturing. Mixed-type wafer defect recognition remains particularly challenging because multiple defect patterns may coexist in a single wafer, leading to substantial spatial overlap and entangled feature representations. To address this problem, we propose DGNet, a dual-collaborative graph network built on a pyramid backbone for mixed-type wafer defect recognition. DGNet consists of two key modules. The Dual-Branch Geometry Learner (DGL) captures multi-scale geometric and topological relationships through dual-branch graph learning, while the Pattern-Aligned Defect Decoupler (PDD) leverages trainable pattern embeddings to disentangle and align mixed-type defect features. Experimental results on the MixedWM38 dataset demonstrate that DGNet achieves an F1 score of 98.93% and shows competitive performance compared with representative state-of-the-art methods. Visualization results further indicate its effectiveness and interpretability in defect localization and feature decoupling.

Keywords—wafer defect pattern recognition, mixed-type defects, graph learning, wafer bin maps

I. INTRODUCTION

Integrated circuit manufacturing begins with the precise processing of high-purity monocrystalline silicon wafers and proceeds through a series of complex fabrication steps to construct nanoscale circuitry. During fabrication, process fluctuations can introduce defects, making wafer testing essential for quality assurance [1]. These test results are commonly summarized as wafer bin maps (WBMs), which reflect wafer-level yield and provide important clues for identifying process anomalies [2]. Wafer defect pattern recognition (WDPR) aims to identify spatial defect patterns from WBMs, thereby supporting defect diagnosis and yield improvement.

Traditional WDPR relies heavily on manual expert judgment, which is inefficient, costly, and subjective. Deep learning methods have therefore been widely studied for learning discriminative features directly from wafer maps. Although many studies have achieved promising performance in single-type defect recognition [3], practical manufacturing scenarios are often more complex. As technology nodes continue to shrink, multiple failure modes may co-occur on the same

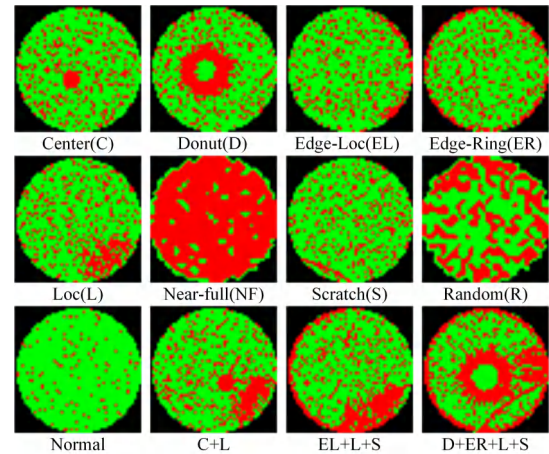


Fig. 1. Examples of single-type and mixed-type wafer defects in the MixedWM38 benchmark. The first two rows show eight single-type defects, while the third row shows normal samples and mixed-type defects containing two, three, or four defect patterns.

wafer. As shown in Fig. 1, mixed-type defect patterns usually exhibit irregular spatial distributions, severe pattern overlap, and strong feature entanglement, making their recognition more challenging.

Existing methods for mixed-type wafer defect recognition can be broadly categorized into machine learning, CNN, Transformer, and recently emerging generative approaches [4]–[7]. Traditional machine learning methods usually rely on handcrafted features and preprocessing, which limits their adaptability to complex mixed defects. CNN methods, such as DCNet [5] and segmentation-based methods [8], improve recognition performance by extracting hierarchical local features or incorporating defect region information. However, their grid-based representations and fixed convolutional operations make it difficult to explicitly model irregular and long-range defect relationships. Transformer methods, represented by MSF-Trans [6], further capture global dependencies through self-attention, but they still lack explicit modeling of spatial adjacency and geometric connectivity. More recently, diffusion methods such as WMDiff [7] have been explored

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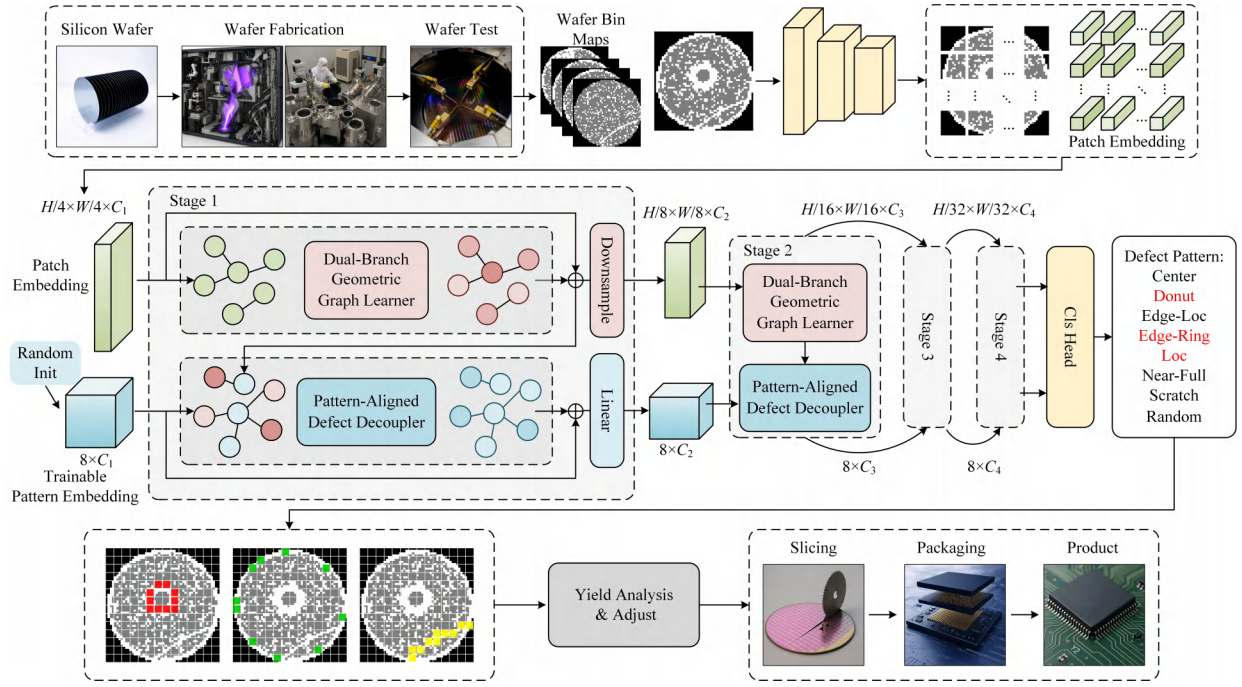


Fig. 2. Overview of the proposed DGNet framework for mixed-type wafer defect recognition. Wafer bin maps are first converted into patch embeddings, which are processed together with trainable pattern embeddings in the pyramid-based dual-collaborative architecture. As the network depth increases, the spatial resolution is progressively reduced while the channel capacity is expanded.

for wafer defect recognition, offering enhanced feature representation capability. Nevertheless, existing methods still face limitations in modeling irregular defect topology and decoupling highly entangled features of mixed defects. Graph neural networks (GNNs) provide a natural alternative by representing image regions or feature tokens as graph nodes and modeling their relationships as edges, enabling explicit representation of discrete, irregular, and highly coupled spatial patterns.

In this paper, we propose a Dual-Collaborative Graph Network (DGNet) for mixed-type wafer defect recognition. Unlike previous grid-based approaches, DGNet reformulates mixed-type defect recognition as a graph classification problem and builds a unified framework for feature learning, pattern correspondence modeling, and multi-label prediction. Built on a pyramid backbone, DGNet consists of three main components. The Dual-Branch Geometry Learner (DGL) captures complex geometric and topological relationships in complementary feature subspaces. The Pattern-Aligned Defect Decoupler (PDD) associates visual features with trainable defect pattern embeddings, enabling category-specific feature alignment and reducing interference among overlapping patterns. Finally, the classification head integrates the decoupled pattern responses to predict the composition of mixed-type defects.

The main contributions are summarized as follows:

- We propose DGNet, a dual-collaborative graph framework that formulates mixed-type wafer defect recognition as a graph classification problem and provides a unified solution for mixed-defect identification.
- We design DGL to capture complex geometric and topological

relationships through dual-branch dynamic graph learning, thereby enhancing the representation of irregular defect structures.

- We introduce PDD to establish explicit correspondence between visual defect features and semantic defect patterns, alleviating feature entanglement and improving interpretability.
- Experimental results on the MixedWM38 dataset demonstrate that DGNet achieves competitive performance compared with representative state-of-the-art methods, while visualization results suggest its potential for defect localization and interpretable mixed-defect analysis.

II. METHODOLOGY

A. Overall Framework

The overall architecture of the proposed framework is illustrated in Fig. 2. First, WBM acquired from wafer testing are processed by three convolutional layers to generate patch embeddings. These patch embeddings, together with randomly initialized trainable pattern embeddings, are fed into the network for defect pattern recognition.

To support deep hierarchical representation learning and multi-scale feature extraction, we adopt the Pyramid ViG-S architecture [9] as the backbone. The backbone consists of four stages, with channel widths C_1 , C_2 , C_3 , and C_4 set to 80, 160, 400, and 640, respectively. The number of basic ViG blocks in the four stages is 2, 2, 6, and 2, respectively.

Building on this backbone, DGNet introduces two key modifications. The original basic ViG blocks are replaced with

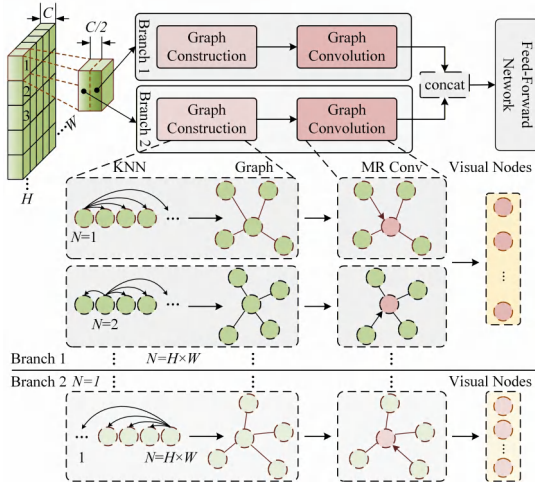


Fig. 3. Illustration of the Dual-Branch Geometry Learner. Here, H and W denote the height and width of the feature map, and MR Conv denotes Max-Relative Graph Convolution.

DGL modules to enhance geometric feature learning, and a PDD module is integrated at the end of each stage to decouple mixed-defect features and reduce feature entanglement.

B. Dual-Branch Geometry Learner (DGL)

The DGL is designed to capture complex defect geometries through dynamic graph learning in two complementary feature subspaces. Given an input feature map $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$, we first split it along the channel dimension into two feature subsets, $\mathbf{X}_1, \mathbf{X}_2 \in \mathbb{R}^{\frac{C}{2} \times H \times W}$.

As shown in Fig. 3, for each branch $b \in \{1, 2\}$, the spatial feature vector $\mathbf{x}_i^{(b)}$ at each position is treated as a graph node. The neighbor set $\mathcal{N}_i^{(b)}$ of each node is dynamically constructed by selecting the k nearest nodes in the feature space according to the Euclidean distance:

$$\mathcal{N}_i^{(b)} = \text{KNN}_{j \neq i} \left(k, \|\mathbf{x}_j^{(b)} - \mathbf{x}_i^{(b)}\|_2 \right), \quad (1)$$

where $\|\mathbf{x}_j^{(b)} - \mathbf{x}_i^{(b)}\|_2$ denotes the Euclidean norm. Based on the constructed graph, node features are updated using Max-Relative Graph Convolution [10], which aggregates the most salient relative differences between each node and its neighbors:

$$\mathbf{x}_i'^{(b)} = \text{Concat} \left(\mathbf{x}_i^{(b)}, \max_{j \in \mathcal{N}_i^{(b)}} (\mathbf{x}_j^{(b)} - \mathbf{x}_i^{(b)}) \right). \quad (2)$$

The updated features from the two branches are then concatenated as $\mathbf{X}' = [\mathbf{X}'_1; \mathbf{X}'_2]$. A feed-forward network (FFN) further projects and fuses the concatenated features, and a residual connection is applied to obtain the final output:

$$\mathbf{Y} = \text{FFN}(\mathbf{X}') + \mathbf{X}. \quad (3)$$

Through this dynamic dual-branch graph learning process, DGL adaptively models geometric relationships in complementary feature subspaces and enhances the representation of non-local spatial dependencies. This capability is particularly

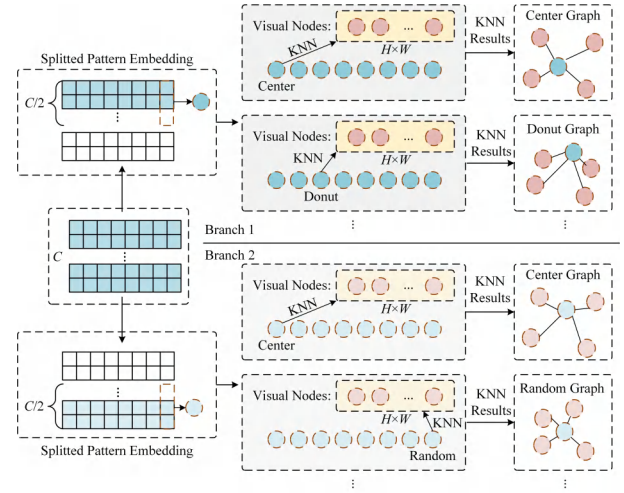


Fig. 4. Illustration of the graph construction process in the Pattern-Aligned Defect Decoupler. Unlike DGL, PDD constructs heterogeneous graphs between semantic pattern embeddings and visual nodes.

beneficial for recognizing mixed-type defects with complex and overlapping spatial structures.

C. Pattern-Aligned Defect Decoupler (PDD)

The PDD module adopts a dual-branch architecture similar to that of DGL, but constructs heterogeneous graphs between semantic pattern embeddings and visual features. The subsequent graph convolution and fusion operations follow a procedure similar to that of DGL. Therefore, Fig. 4 focuses on the graph construction process of PDD.

PDD takes as input the feature map $\mathbf{Y} \in \mathbb{R}^{C \times H \times W}$ from the preceding module and a set of trainable pattern embeddings $\mathbf{P} \in \mathbb{R}^{C \times K}$, where K denotes the number of basic defect patterns. Both $\mathbf{Y}$ and $\mathbf{P}$ are split along the channel dimension into two branches, each with $\frac{C}{2}$ channels. For each branch $b \in \{1, 2\}$, the pattern embeddings are treated as semantic nodes, while the spatial feature vectors are treated as visual nodes. For each semantic node $\mathbf{p}_i^{(b)}$, its neighbor set $\mathcal{N}_i^{(b)}$ is dynamically constructed by selecting the k nearest visual nodes based on Euclidean distance:

$$\mathcal{N}_i^{(b)} = \text{KNN}_j \left(k, \|\mathbf{p}_i^{(b)} - \mathbf{y}_j^{(b)}\|_2 \right), \quad (4)$$

where $\mathbf{y}_j^{(b)}$ denotes the j -th visual node in branch b . The semantic nodes are then updated using Max-Relative Graph Convolution, which aggregates the maximum relative differences between each semantic node and its visual neighbors:

$$\mathbf{p}_i'^{(b)} = \text{Concat} \left(\mathbf{p}_i^{(b)}, \max_{j \in \mathcal{N}_i^{(b)}} (\mathbf{y}_j^{(b)} - \mathbf{p}_i^{(b)}) \right). \quad (5)$$

After graph convolution in both branches, the updated pattern embeddings $\mathbf{P}'_1$ and $\mathbf{P}'_2$ are concatenated and refined by an FFN with a residual connection:

$$\mathbf{P}' = \text{FFN}([\mathbf{P}'_1; \mathbf{P}'_2]) + \mathbf{P}. \quad (6)$$

TABLE I
ABLATION STUDY OF DGNET ON THE MIXEDWM38 DATASET (%)

	submodules			Precision	Recall	F1 Score
	+ DGL	+ PDD	+ Pre-trained			
Baseline	-	-	-	96.92	96.87	96.86
DGNet	-	✓	✓	98.69	98.68	98.68
	✓	-	✓	98.70	98.68	98.69
	✓	✓	-	97.97	97.94	97.94
	✓	✓	✓	98.94	98.93	98.93

Through this dual-branch graph interaction and fusion process, PDD models pattern–feature correlations in complementary feature subspaces and aligns trainable defect pattern embeddings with visual representations.

D. Classification Head

To align decoupled defect geometry features with their corresponding defect patterns, the classification head introduces a feature–pattern alignment mechanism. In the visual branch, a matrix-based alignment operation establishes a one-to-one correspondence between visual features and defect categories, encouraging each category’s logit to be derived from its corresponding visual representation. The aligned visual logits are then fused with the logits from the pattern-embedding branch to produce the final prediction scores. During inference, a Sigmoid activation function is applied, and a fixed classification threshold of 0.62 is used to obtain the final binary predictions.

E. Loss Function

To address multi-label classification and class imbalance in mixed-defect scenarios, we adopt a composite loss function. The total loss L_{total} is defined as the weighted sum of two components:

$$L_{\text{total}} = L_{\text{BCE-LS}} + w \cdot L_{\text{ASL}}, \quad (7)$$

where $L_{\text{BCE-LS}}$ denotes the binary cross-entropy loss with label smoothing [11], L_{ASL} denotes the asymmetric loss [12], and w is the weighting coefficient. The former improves generalization by softening hard labels, while the latter mitigates the imbalance between positive and negative labels in multi-label classification.

III. EXPERIMENTAL RESULTS

A. Experimental Setting

To evaluate the recognition of complex mixed-type defect patterns, this study uses the MixedWM38 benchmark [5], which is derived from real production scenarios. The dataset contains 38 categories constructed from eight representative defect types: Center (C), Donut (D), Loc (L), Edge-Loc (EL), Edge-Ring (ER), Near-Full (NF), Scratch (S), and Random (R). These defect types are not only visual patterns but are also closely associated with specific manufacturing process steps and defect formation mechanisms [2].

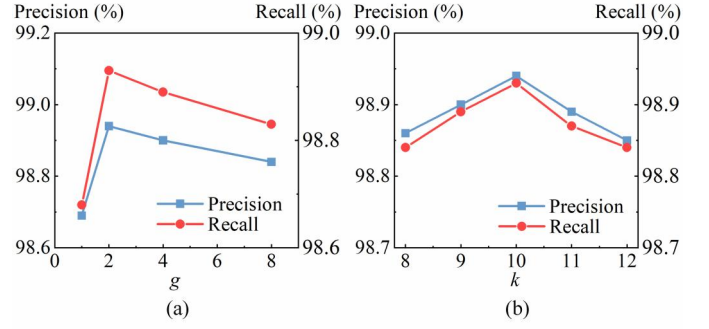


Fig. 5. Sensitivity analysis of key hyperparameters: (a) branch number g and (b) neighbor count k in KNN-based graph construction.

The MixedWM38 dataset was split into training and test sets with a fixed 8:2 ratio, following a commonly used evaluation setting in prior MixedWM38 studies. All experiments were conducted on a distributed computing platform equipped with 8 NVIDIA GeForce RTX 4080 GPUs, running Linux and PyTorch. We used the AdamW optimizer with an initial learning rate of 0.0001, together with weight decay and gradient clipping to improve training stability. The model was trained for 50 epochs with a batch size of 32 and initialized with ImageNet-pretrained weights.

During preprocessing, the raw 52×52 single-channel wafer images were upsampled to a resolution of 256×256 using nearest-neighbor interpolation and converted into a three-channel RGB format to match the input requirements of ImageNet pretrained models. Data augmentation techniques, including geometric transformations, intensity adjustments, and sample-mixing strategies, were further applied to improve model generalization.

B. Ablation study and Hyperparameter Analysis

To evaluate the effectiveness of the proposed modules and the contribution of pre-training, we conducted ablation experiments, as shown in Table I. A standard Pyramid ViG-S network was used as the baseline, achieving an F1 score of 96.86%. The complete DGNet, which integrates DGL, PDD, and pre-trained initialization, improved the F1 score to 98.93%, yielding a gain of more than 2 percentage points. To further analyze the contribution of each component, we removed DGL, PDD, and the pre-trained weights from the complete model, respectively. Removing PDD reduced the F1 score to 98.69%, while removing DGL led to a comparable decrease to 98.68%, indicating that both modules contribute to performance improvement. When the pre-trained weights were removed, the F1 score decreased to 97.94%, even with DGL and PDD retained. Although wafer maps differ substantially from natural images, ImageNet pre-training still provides useful low-level visual priors and spatial-structure representations. These generic features offer an effective initialization and help improve model generalization. Overall, the ablation results show that DGL, PDD, and pre-training complement each other in enhancing the performance of DGNet.

TABLE II
PERFORMANCE COMPARISON BETWEEN DGNET AND REPRESENTATIVE STATE-OF-THE-ART METHODS ON THE MIXEDWM38 DATASET.

Pattern	MSF-Trans [6]			Segmentation-based [8]			WMDiff [7]			DGNet (Ours)		
	Precision	Recall	F1 Score	Precision	Recall	F1 Score	Precision	Recall	F1 Score	Precision	Recall	F1 Score
Normal	0.974	1.000	0.987	1.000	1.000	1.000	1.000	1.000	1.000	0.997	1.000	0.998
Center	0.936	0.973	0.954	1.000	0.988	0.994	0.982	1.000	0.991	1.000	0.990	0.995
Donut	0.967	0.973	0.970	0.985	0.990	0.987	0.962	0.995	0.978	0.987	1.000	0.993
Edge-Loc	0.976	0.937	0.956	0.989	1.000	0.994	0.989	0.989	0.989	1.000	0.990	0.995
Edge-Ring	0.938	0.963	0.950	0.980	1.000	0.990	0.975	0.996	0.985	0.987	0.997	0.992
Loc	0.961	0.977	0.969	0.990	0.981	0.986	0.967	0.986	0.977	0.990	0.997	0.993
Near-Full	0.992	0.950	0.971	0.958	1.000	0.979	0.971	0.971	0.971	0.980	1.000	0.990
Scratch	0.954	0.970	0.962	0.981	1.000	0.990	0.984	1.000	0.992	0.987	0.997	0.992
Random	0.759	0.911	0.828	1.000	0.988	0.994	0.994	0.994	0.994	1.000	0.979	0.990
C+EL	0.954	0.960	0.957	0.990	0.990	0.990	0.979	0.979	0.979	0.974	0.997	0.985
C+ER	0.951	0.977	0.964	0.981	0.981	0.981	0.946	1.000	0.972	0.984	1.000	0.992
C+L	0.960	0.963	0.961	0.990	0.933	0.961	0.963	0.979	0.971	0.993	0.993	0.993
C+S	0.930	0.973	0.951	0.960	0.982	0.970	0.978	0.994	0.986	0.990	0.997	0.993
D+EL	0.944	0.957	0.950	0.980	0.985	0.982	0.981	0.976	0.979	0.997	0.983	0.990
D+ER	0.917	0.960	0.938	1.000	0.969	0.984	0.984	0.995	0.989	0.993	0.997	0.995
D+L	0.935	0.953	0.944	0.984	0.949	0.966	0.972	0.972	0.972	1.000	0.970	0.985
D+S	0.956	0.950	0.953	0.959	0.991	0.975	0.946	0.985	0.965	0.984	1.000	0.992
EL+L	0.912	0.970	0.940	0.984	0.925	0.954	0.972	0.983	0.978	0.990	1.000	0.995
EL+S	0.972	0.933	0.952	0.983	0.973	0.978	0.995	0.979	0.987	0.983	0.977	0.980
ER+L	0.927	0.967	0.947	0.986	0.931	0.958	0.967	0.958	0.963	0.993	0.973	0.983
ER+S	0.960	0.960	0.960	0.972	0.995	0.984	0.971	0.995	0.983	0.977	1.000	0.988
L+S	0.963	0.963	0.963	0.983	0.944	0.963	0.965	0.960	0.962	0.993	0.980	0.987
C+EL+L	0.979	0.953	0.966	0.980	0.903	0.940	0.975	0.934	0.954	0.990	0.970	0.980
C+EL+S	0.988	0.977	0.982	0.965	0.965	0.965	0.995	0.995	0.995	0.997	0.997	0.997
C+ER+L	0.873	0.940	0.905	0.970	0.956	0.963	0.947	0.957	0.952	0.993	0.977	0.985
C+ER+S	0.973	0.977	0.975	0.975	0.970	0.972	0.971	0.990	0.981	0.987	0.997	0.992
C+L+S	0.983	0.937	0.959	0.942	0.985	0.963	0.983	0.938	0.960	0.983	0.990	0.987
D+EL+L	0.937	0.937	0.937	0.990	0.944	0.966	1.000	0.962	0.981	1.000	0.997	0.998
D+EL+S	0.951	0.903	0.926	0.958	0.976	0.967	0.977	0.976	0.976	0.987	0.977	0.982
D+ER+L	0.970	0.957	0.963	0.984	0.926	0.954	0.976	1.000	0.988	0.990	0.997	0.993
D+ER+S	0.884	0.940	0.911	0.948	0.980	0.964	0.970	0.995	0.982	0.980	1.000	0.990
D+L+S	0.968	0.917	0.942	0.950	0.974	0.962	0.971	0.939	0.955	0.980	0.983	0.982
EL+L+S	0.978	0.910	0.943	0.920	0.964	0.941	0.985	0.930	0.957	0.997	0.970	0.983
ER+L+S	0.935	0.917	0.926	0.952	0.952	0.952	0.962	0.941	0.951	0.977	0.990	0.983
C+L+EL+S	0.959	0.930	0.944	0.913	0.931	0.922	0.969	0.945	0.957	1.000	0.963	0.981
C+L+ER+S	0.932	0.863	0.896	0.958	0.968	0.963	0.974	0.949	0.961	0.970	0.990	0.980
D+L+EL+S	0.965	0.910	0.937	0.946	0.946	0.946	0.989	0.920	0.953	0.997	0.987	0.992
D+L+ER+S	0.944	0.953	0.948	0.959	0.964	0.962	0.975	0.995	0.985	0.990	0.993	0.992
single	0.940	0.962	0.950	0.987	0.994	0.990	0.980	0.992	0.986	0.992	0.994	0.993
two-mixed	0.945	0.960	0.952	0.981	0.965	0.973	0.971	0.981	0.976	0.989	0.990	0.989
three-mixed	0.952	0.939	0.945	0.961	0.958	0.959	0.976	0.963	0.969	0.988	0.987	0.988
four-mixed	0.950	0.914	0.931	0.944	0.952	0.948	0.977	0.952	0.964	0.989	0.983	0.986
Average	0.946	0.949	0.948	0.972	0.968	0.970	0.975	0.975	0.975	0.989	0.989	0.989

Fig. 5 compares the effects of key hyperparameters on model performance. To validate the dual-branch design, we evaluated models with different numbers of branches. When the branch number g was set to 2, the model achieved the highest F1 score of 98.93%. Although more branches may enable finer-grained feature separation, they may also disrupt intrinsic feature relationships and introduce additional computational overhead, thereby degrading performance. Second, we investigated the effect of the neighbor count k in KNN-based graph construction. As k increased from 8 to 12, the F1 score first increased and then decreased, reaching its peak of 98.93% at $k = 10$. This trend reflects a trade-off in neighborhood

selection. A small k leads to sparse graph connections, limiting graph convolution in capturing sufficient geometric structures. In contrast, a large k introduces redundant connections, which may increase noise interference and hinder the identification of discriminative defect patterns in wafer maps.

C. Compared with State-of-the-Art Methods

The proposed DGNet is compared with representative state-of-the-art (SOTA) methods reported on the MixedWM38 dataset, including MSF-Trans [6], segmentation-based [8], and WMDiff [7]. As shown in Table II, DGNet achieves the best overall performance, with average precision, recall, and F1 score all reaching 0.989. Its advantage becomes more

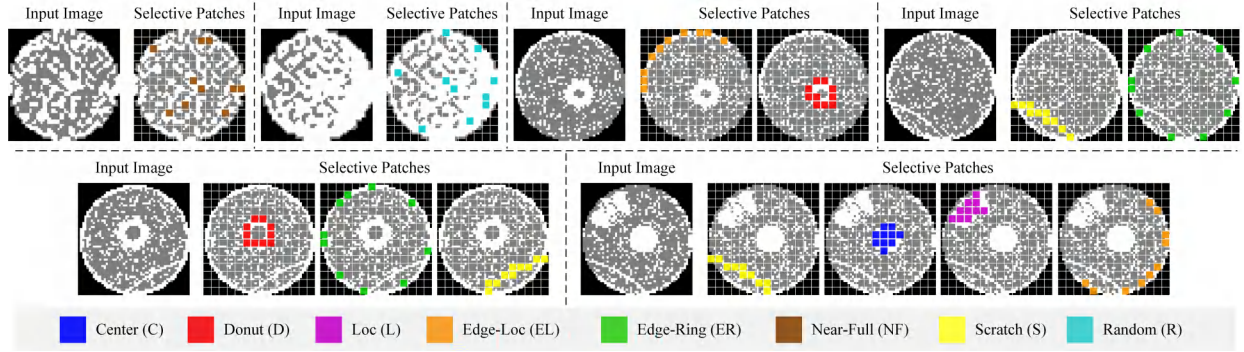


Fig. 6. Visualization of input wafer maps and the top-10 patches selected for each defect type under different defect-complexity levels. NF and R typically appear as isolated instances.

evident in complex mixed-defect scenarios. DGNet obtains F1 scores of 0.989, 0.988, and 0.986 for two-mixed, three-mixed, and four-mixed defects. In the most challenging four-mixed setting, DGNet outperforms WMDiff, segmentation-based, and MSF-Trans by 0.022, 0.038, and 0.055 in F1 score, respectively. These results indicate that DGNet provides strong robustness to severe spatial overlap and feature ambiguity, benefiting from DGL's geometric relationship modeling and PDD's pattern-specific feature decoupling.

In addition, DGNet achieves an F1 score of 98.93% with 5.19 G FLOPs and 33.26 M parameters. Since FLOPs do not fully reflect the memory-bandwidth overhead of dynamic KNN and many baselines lack matched latency reports, lightweight acceleration such as approximate KNN, pruning, quantization, and distillation will be explored for deployment.

D. Qualitative Analysis

The decision process of DGNet is further examined by visualizing the patches selected for each defect type, as shown in Fig. 6. Specifically, for each defect-category query, the 10 most relevant patches are localized in the input wafer map. These highlighted regions indicate the spatial evidence used by the model for recognizing both single-type and mixed-type defects, providing interpretable cues for subsequent defect diagnosis and root-cause analysis.

For defect types with clear local structures, such as C, L, and S, the selected patches are well aligned with the corresponding defect regions, even in complex mixed-defect cases. For global or dispersed defects, such as ER and NF, the top-10 patches may not cover all defective regions due to their wide spatial distribution. Nevertheless, the selected patches still capture representative distribution patterns of these defects. These visualization results suggest that DGNet can focus on defect-relevant regions and provide interpretable evidence for mixed-defect recognition.

IV. CONCLUSION

In this paper, we propose DGNet, a dual-collaborative graph framework for mixed-type wafer defect recognition. By integrating DGL and PDD into a pyramid backbone, DGNet models defect geometric topology and promotes semantic-visual

pattern decoupling. Experiments on MixedWM38 achieve an F1 score of 98.93%, and visualizations suggest interpretable defect localization. Future work will extend DGNet to diverse industrial datasets, open-set and few-shot defect discovery, wafer-domain pre-training, and lightweight deployment.

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